

# Integration with Partial Fractions — Solutions

## Warm-up

### Question 1i:

To find the values of  $A$  and  $B$ , we start with the partial fraction decomposition:

$$\begin{aligned}\frac{1}{(x+1)(x+2)} &= \frac{A}{x+1} + \frac{B}{x+2} \\ 1 &= A(x+2) + B(x+1) \\ \text{Let } x = -1 : &\implies 1 = A(-1+2) \\ &\implies A = 1 \\ \text{Let } x = -2 : &\implies 1 = B(-2+1) \\ &\implies B = -1\end{aligned}$$

Thus,  $A = 1$  and  $B = -1$ .

### Solution to Question 1ii:

Using the result from part (i), we substitute the partial fractions into the integral:

$$\begin{aligned}\int \frac{1}{(x+1)(x+2)} dx &= \int \left( \frac{1}{x+1} - \frac{1}{x+2} \right) dx \\ &= \ln|x+1| - \ln|x+2| + C \\ &= \ln \left| \frac{x+1}{x+2} \right| + C\end{aligned}$$

### Solution to Question 2i:

We set up the partial fraction decomposition:

$$\begin{aligned}\frac{3}{(x+3)(x+6)} &= \frac{A}{x+3} + \frac{B}{x+6} \\ 3 &= A(x+6) + B(x+3)\end{aligned}$$

Solving for  $A$  and  $B$ :

$$\begin{aligned}\text{Let } x = -3 : &\implies 3 = A(-3+6) \\ &\implies 3 = 3A \\ &\implies A = 1 \\ \text{Let } x = -6 : &\implies 3 = B(-6+3) \\ &\implies 3 = -3B \\ &\implies B = -1\end{aligned}$$

Thus,  $A = 1$  and  $B = -1$ .

### Solution to Question 2ii:

Using the result from part (i):

$$\begin{aligned}
\int \frac{3}{(x+3)(x+6)} dx &= \int \left( \frac{1}{x+3} - \frac{1}{x+6} \right) dx \\
&= \ln|x+3| - \ln|x+6| + C \\
&= \ln \left| \frac{x+3}{x+6} \right| + C
\end{aligned}$$

**Solution to Question 3i:**

We set up the partial fraction decomposition:

$$\begin{aligned}
\frac{1}{(x+6)(2x+7)} &= \frac{A}{x+6} + \frac{B}{2x+7} \\
1 &= A(2x+7) + B(x+6)
\end{aligned}$$

Solving for  $A$  and  $B$ :

$$\begin{aligned}
\text{Let } x = -6 : &\implies 1 = A(2(-6) + 7) \\
&\implies 1 = A(-5) \\
&\implies A = -\frac{1}{5} \\
\text{Let } x = -\frac{7}{2} : &\implies 1 = B(-\frac{7}{2} + 6) \\
&\implies 1 = B(\frac{5}{2}) \\
&\implies B = \frac{2}{5}
\end{aligned}$$

Thus,  $A = -\frac{1}{5}$  and  $B = \frac{2}{5}$ .

**Solution to Question 3ii:**

Using the result from part (i):

$$\begin{aligned}
\int \frac{1}{(x+6)(2x+7)} dx &= \int \left( \frac{-1/5}{x+6} + \frac{2/5}{2x+7} \right) dx \\
&= -\frac{1}{5} \ln|x+6| + \frac{2}{5} \int \frac{1}{2x+7} dx \\
&= -\frac{1}{5} \ln|x+6| + \frac{2}{5} \cdot \frac{1}{2} \ln|2x+7| + C \\
&= -\frac{1}{5} \ln|x+6| + \frac{1}{5} \ln|2x+7| + C \\
&= \frac{1}{5} \ln \left| \frac{2x+7}{x+6} \right| + C
\end{aligned}$$

**Solution to Question 4i:**

We set up the partial fraction decomposition:

$$\begin{aligned}
\frac{1}{(x+2)(x^2+4)} &= \frac{A}{x+2} + \frac{Bx+C}{x^2+4} \\
1 &= A(x^2+4) + (Bx+C)(x+2)
\end{aligned}$$

Solving for  $A$ :

$$\begin{aligned}\text{Let } x = -2 : & \implies 1 = A((-2)^2 + 4) \\ & \implies 1 = 8A \\ & \implies A = \frac{1}{8}\end{aligned}$$

Expanding and comparing coefficients for  $B$  and  $C$ :

$$\begin{aligned}1 &= \frac{1}{8}(x^2 + 4) + Bx^2 + 2Bx + Cx + 2C \\ 1 &= \left(\frac{1}{8} + B\right)x^2 + (2B + C)x + \left(\frac{1}{2} + 2C\right)\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}\frac{1}{8} + B &= 0 \\ \implies B &= -\frac{1}{8}\end{aligned}$$

Comparing coefficients of  $x^0$  (constants):

$$\begin{aligned}\frac{1}{2} + 2C &= 1 \\ \implies 2C &= \frac{1}{2} \\ \implies C &= \frac{1}{4}\end{aligned}$$

Thus,  $A = \frac{1}{8}, B = -\frac{1}{8}, C = \frac{1}{4}$ .

#### **Solution to Question 4ii:**

Using the result from part (i):

$$\begin{aligned}\int \frac{1}{(x+2)(x^2+4)} dx &= \int \left( \frac{1/8}{x+2} + \frac{-x/8 + 1/4}{x^2+4} \right) dx \\ &= \frac{1}{8} \int \frac{1}{x+2} dx - \frac{1}{8} \int \frac{x}{x^2+4} dx + \frac{1}{4} \int \frac{1}{x^2+4} dx \\ &= \frac{1}{8} \ln|x+2| - \frac{1}{8} \cdot \frac{1}{2} \ln(x^2+4) + \frac{1}{4} \cdot \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C \\ &= \frac{1}{8} \ln|x+2| - \frac{1}{16} \ln(x^2+4) + \frac{1}{8} \tan^{-1}\left(\frac{x}{2}\right) + C\end{aligned}$$

#### **Solution to Question 5i:**

We set up the partial fraction decomposition:

$$\begin{aligned}\frac{2x+5}{(x+2)(x^2+4)} &= \frac{A}{x+2} + \frac{Bx+C}{x^2+4} \\ 2x+5 &= A(x^2+4) + (Bx+C)(x+2)\end{aligned}$$

Solving for  $A$ :

$$\begin{aligned}
\text{Let } x = -2 : & \implies 2(-2) + 5 = A((-2)^2 + 4) \\
& \implies 1 = 8A \\
& \implies A = \frac{1}{8}
\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}
2x + 5 &= \frac{1}{8}x^2 + \frac{1}{2} + Bx^2 + 2Bx + Cx + 2C \\
2x + 5 &= \left(\frac{1}{8} + B\right)x^2 + (2B + C)x + \left(\frac{1}{2} + 2C\right)
\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}
\frac{1}{8} + B &= 0 \\
\implies B &= -\frac{1}{8}
\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}
\frac{1}{2} + 2C &= 5 \\
\implies 2C &= 4.5 = \frac{9}{2} \\
\implies C &= \frac{9}{4}
\end{aligned}$$

Thus,  $A = \frac{1}{8}, B = -\frac{1}{8}, C = \frac{9}{4}$ .

### Solution to Question 5ii:

Using the result from part (i):

$$\begin{aligned}
\int \frac{2x + 5}{(x + 2)(x^2 + 4)} dx &= \int \left( \frac{1/8}{x + 2} + \frac{-x/8 + 9/4}{x^2 + 4} \right) dx \\
&= \frac{1}{8} \int \frac{1}{x + 2} dx - \frac{1}{8} \int \frac{x}{x^2 + 4} dx + \frac{9}{4} \int \frac{1}{x^2 + 4} dx \\
&= \frac{1}{8} \ln |x + 2| - \frac{1}{16} \ln(x^2 + 4) + \frac{9}{4} \cdot \frac{1}{2} \tan^{-1} \left( \frac{x}{2} \right) + C \\
&= \frac{1}{8} \ln |x + 2| - \frac{1}{16} \ln(x^2 + 4) + \frac{9}{8} \tan^{-1} \left( \frac{x}{2} \right) + C
\end{aligned}$$

## Skill-building

### Solution to Question 1:

First, we decompose the integrand into partial fractions:

$$\begin{aligned}
\frac{1}{(x + 2)(x - 2)} &= \frac{A}{x + 2} + \frac{B}{x - 2} \\
1 &= A(x - 2) + B(x + 2)
\end{aligned}$$

Solving for  $A$  and  $B$ :

$$\begin{aligned}\text{Let } x = 2 : & \implies 1 = B(4) \\ & \implies B = \frac{1}{4} \\ \text{Let } x = -2 : & \implies 1 = A(-4) \\ & \implies A = -\frac{1}{4}\end{aligned}$$

Now, integrate:

$$\begin{aligned}\int \frac{1}{(x+2)(x-2)} dx &= \int \left( \frac{-1/4}{x+2} + \frac{1/4}{x-2} \right) dx \\ &= -\frac{1}{4} \ln |x+2| + \frac{1}{4} \ln |x-2| + C \\ &= \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + C\end{aligned}$$

### **Solution to Question 2:**

Decompose the integrand into partial fractions:

$$\begin{aligned}\frac{1}{(2x+1)(2x-1)} &= \frac{A}{2x+1} + \frac{B}{2x-1} \\ 1 &= A(2x-1) + B(2x+1)\end{aligned}$$

Solving for  $A$  and  $B$ :

$$\begin{aligned}\text{Let } x = 1/2 : & \implies 1 = B(2) \\ & \implies B = \frac{1}{2} \\ \text{Let } x = -1/2 : & \implies 1 = A(-2) \\ & \implies A = -\frac{1}{2}\end{aligned}$$

Now, integrate with limits:

$$\begin{aligned}\int_1^2 \frac{1}{(2x+1)(2x-1)} dx &= \int_1^2 \left( \frac{-1/2}{2x+1} + \frac{1/2}{2x-1} \right) dx \\ &= \left[ -\frac{1}{2} \cdot \frac{1}{2} \ln |2x+1| + \frac{1}{2} \cdot \frac{1}{2} \ln |2x-1| \right]_1^2 \\ &= \left[ \frac{1}{4} \ln \left| \frac{2x-1}{2x+1} \right| \right]_1^2 \\ &= \frac{1}{4} \left( \ln \left| \frac{3}{5} \right| - \ln \left| \frac{1}{3} \right| \right) \\ &= \frac{1}{4} \ln \left| \frac{3/5}{1/3} \right| = \frac{1}{4} \ln \left( \frac{9}{5} \right)\end{aligned}$$

### **Solution to Question 3:**

Decompose the integrand into partial fractions:

$$\frac{x}{(2x+3)(x+4)} = \frac{A}{2x+3} + \frac{B}{x+4}$$

$$x = A(x+4) + B(2x+3)$$

Solving for  $A$  and  $B$ :

$$\begin{aligned}\text{Let } x = -4 : & \implies -4 = B(2(-4) + 3) \\ & \implies -4 = -5B \\ & \implies B = \frac{4}{5} \\ \text{Let } x = -3/2 : & \implies -3/2 = A(-3/2 + 4) \\ & \implies -3/2 = A(5/2) \\ & \implies A = -\frac{3}{5}\end{aligned}$$

Now, integrate:

$$\begin{aligned}\int \frac{x}{(2x+3)(x+4)} dx &= \int \left( \frac{-3/5}{2x+3} + \frac{4/5}{x+4} \right) dx \\ &= -\frac{3}{5} \cdot \frac{1}{2} \ln |2x+3| + \frac{4}{5} \ln |x+4| + C \\ &= -\frac{3}{10} \ln |2x+3| + \frac{4}{5} \ln |x+4| + C\end{aligned}$$

#### **Solution to Question 4:**

Decompose the integrand into partial fractions:

$$\frac{x+5}{(4x+1)(3x-1)} = \frac{A}{4x+1} + \frac{B}{3x-1}$$

$$x+5 = A(3x-1) + B(4x+1)$$

Solving for  $A$  and  $B$ :

$$\begin{aligned}\text{Let } x = 1/3 : & \implies 1/3 + 5 = B(4/3 + 1) \\ & \implies 16/3 = B(7/3) \\ & \implies B = \frac{16}{7} \\ \text{Let } x = -1/4 : & \implies -1/4 + 5 = A(3(-1/4) - 1) \\ & \implies 19/4 = A(-7/4) \\ & \implies A = -\frac{19}{7}\end{aligned}$$

Now, integrate:

$$\begin{aligned}\int \frac{x+5}{(4x+1)(3x-1)} dx &= \int \left( \frac{-19/7}{4x+1} + \frac{16/7}{3x-1} \right) dx \\ &= -\frac{19}{7} \cdot \frac{1}{4} \ln |4x+1| + \frac{16}{7} \cdot \frac{1}{3} \ln |3x-1| + C \\ &= -\frac{19}{28} \ln |4x+1| + \frac{16}{21} \ln |3x-1| + C\end{aligned}$$

**Solution to Question 5:**

Decompose the integrand into partial fractions:

$$\begin{aligned}\frac{1}{(x^2 + 1)(x - 1)} &= \frac{A}{x - 1} + \frac{Bx + C}{x^2 + 1} \\ 1 &= A(x^2 + 1) + (Bx + C)(x - 1)\end{aligned}$$

Solving for  $A$ :

$$\begin{aligned}\text{Let } x = 1 : &\implies 1 = A(1^2 + 1) \\ &\implies 1 = 2A \\ &\implies A = \frac{1}{2}\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}1 &= \frac{1}{2}x^2 + \frac{1}{2} + Bx^2 - Bx + Cx - C \\ 1 &= \left(\frac{1}{2} + B\right)x^2 + (C - B)x + \left(\frac{1}{2} - C\right)\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}\frac{1}{2} + B &= 0 \\ \implies B &= -\frac{1}{2}\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}\frac{1}{2} - C &= 1 \\ \implies C &= -\frac{1}{2}\end{aligned}$$

Now, integrate:

$$\begin{aligned}\int \frac{1}{(x^2 + 1)(x - 1)} dx &= \int \left( \frac{1/2}{x - 1} + \frac{-x/2 - 1/2}{x^2 + 1} \right) dx \\ &= \frac{1}{2} \int \frac{1}{x - 1} dx - \frac{1}{2} \int \frac{x}{x^2 + 1} dx - \frac{1}{2} \int \frac{1}{x^2 + 1} dx \\ &= \frac{1}{2} \ln|x - 1| - \frac{1}{2} \cdot \frac{1}{2} \ln(x^2 + 1) - \frac{1}{2} \tan^{-1} x + C \\ &= \frac{1}{2} \ln|x - 1| - \frac{1}{4} \ln(x^2 + 1) - \frac{1}{2} \tan^{-1} x + C\end{aligned}$$

**Solution to Question 6:**

First, simplify the integrand. Note that  $2x^2 + 4 = 2(x^2 + 2)$ :

$$\frac{x + 1}{2(x^2 + 2)(x - 3)} = \frac{1}{2} \left( \frac{x + 1}{(x^2 + 2)(x - 3)} \right)$$

Decompose the inner fraction into partial fractions:

$$\frac{x+1}{(x^2+2)(x-3)} = \frac{A}{x-3} + \frac{Bx+C}{x^2+2}$$

$$x+1 = A(x^2+2) + (Bx+C)(x-3)$$

Solving for A:

$$\begin{aligned}\text{Let } x = 3 : &\implies 3+1 = A(3^2+2) \\ &\implies 4 = 11A \\ &\implies A = \frac{4}{11}\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}x+1 &= \frac{4}{11}x^2 + \frac{8}{11} + Bx^2 - 3Bx + Cx - 3C \\ x+1 &= \left(\frac{4}{11} + B\right)x^2 + (C - 3B)x + \left(\frac{8}{11} - 3C\right)\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}\frac{4}{11} + B &= 0 \\ \implies B &= -\frac{4}{11}\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}\frac{8}{11} - 3C &= 1 \\ \implies -3C &= 1 - \frac{8}{11} = \frac{3}{11} \\ \implies C &= -\frac{1}{11}\end{aligned}$$

Now, integrate the whole expression (including the 1/2 factor):

$$\begin{aligned}\int_1^2 \frac{x+1}{2(x^2+2)(x-3)} dx &= \frac{1}{2} \int_1^2 \left( \frac{4/11}{x-3} + \frac{-4x/11 - 1/11}{x^2+2} \right) dx \\ &= \frac{1}{22} \int_1^2 \left( \frac{4}{x-3} - \frac{4x+1}{x^2+2} \right) dx \\ &= \frac{1}{22} \left[ 4 \ln|x-3| - 2 \ln(x^2+2) - \frac{1}{\sqrt{2}} \tan^{-1} \left( \frac{x}{\sqrt{2}} \right) \right]_1^2 \\ &= \frac{1}{22} \left( (4 \ln|-1| - 2 \ln 6 - \frac{1}{\sqrt{2}} \tan^{-1} \frac{2}{\sqrt{2}}) - (4 \ln|-2| - 2 \ln 3 - \frac{1}{\sqrt{2}} \tan^{-1} \frac{1}{\sqrt{2}}) \right) \\ &= \frac{1}{22} \left( -2 \ln 6 - \frac{1}{\sqrt{2}} \tan^{-1} \sqrt{2} - 4 \ln 2 + 2 \ln 3 + \frac{1}{\sqrt{2}} \tan^{-1} \frac{1}{\sqrt{2}} \right) \\ &= \frac{1}{22} \left( -2 \ln 2 - 4 \ln 2 - \frac{1}{\sqrt{2}} (\tan^{-1} \sqrt{2} - \tan^{-1} \frac{1}{\sqrt{2}}) \right) \\ &= \frac{1}{22} \left( -6 \ln 2 - \frac{1}{\sqrt{2}} (\tan^{-1} \sqrt{2} - \tan^{-1} \frac{1}{\sqrt{2}}) \right)\end{aligned}$$

**Solution to Question 7:**

Decompose the integrand into partial fractions:

$$\frac{x^2 + 3x + 4}{(x^2 + 9)(x + 1)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 9}$$

$$x^2 + 3x + 4 = A(x^2 + 9) + (Bx + C)(x + 1)$$

Solving for  $A$ :

$$\begin{aligned}\text{Let } x = -1 : & \implies (-1)^2 + 3(-1) + 4 = A((-1)^2 + 9) \\ & \implies 2 = 10A \\ & \implies A = \frac{1}{5}\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}x^2 + 3x + 4 &= \frac{1}{5}x^2 + \frac{9}{5} + Bx^2 + Bx + Cx + C \\ x^2 + 3x + 4 &= \left(\frac{1}{5} + B\right)x^2 + (B + C)x + \left(\frac{9}{5} + C\right)\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}\frac{1}{5} + B &= 1 \\ \implies B &= \frac{4}{5}\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}\frac{9}{5} + C &= 4 \\ \implies C &= 4 - \frac{9}{5} = \frac{11}{5}\end{aligned}$$

Now, integrate from 0 to 1:

$$\begin{aligned}\int_0^1 \frac{x^2 + 3x + 4}{(x^2 + 9)(x + 1)} dx &= \int_0^1 \left( \frac{1/5}{x + 1} + \frac{4x/5 + 11/5}{x^2 + 9} \right) dx \\ &= \frac{1}{5} \int_0^1 \frac{1}{x + 1} dx + \frac{4}{5} \int_0^1 \frac{x}{x^2 + 9} dx + \frac{11}{5} \int_0^1 \frac{1}{x^2 + 9} dx \\ &= \left[ \frac{1}{5} \ln |x + 1| + \frac{4}{5} \cdot \frac{1}{2} \ln(x^2 + 9) + \frac{11}{5} \cdot \frac{1}{3} \tan^{-1} \left( \frac{x}{3} \right) \right]_0^1 \\ &= \left[ \frac{1}{5} \ln |x + 1| + \frac{2}{5} \ln(x^2 + 9) + \frac{11}{15} \tan^{-1} \left( \frac{x}{3} \right) \right]_0^1 \\ &= \left( \frac{1}{5} \ln 2 + \frac{2}{5} \ln 10 + \frac{11}{15} \tan^{-1} \frac{1}{3} \right) - \left( \frac{1}{5} \ln 1 + \frac{2}{5} \ln 9 + \frac{11}{15} \tan^{-1} 0 \right) \\ &= \frac{1}{5} \ln 2 + \frac{2}{5} \ln 10 - \frac{2}{5} \ln 9 + \frac{11}{15} \tan^{-1} \frac{1}{3} \\ &= \frac{1}{5} \ln 2 + \frac{2}{5} \ln \left( \frac{10}{9} \right) + \frac{11}{15} \tan^{-1} \frac{1}{3}\end{aligned}$$

## Easier Exam Questions

### Solution to Question 1i:

We set up the partial fraction decomposition:

$$\begin{aligned}\frac{6x^2 + 7x - 3}{(x - 5)(x^2 + 1)} &= \frac{A}{x - 5} + \frac{Bx + C}{x^2 + 1} \\ 6x^2 + 7x - 3 &= A(x^2 + 1) + (Bx + C)(x - 5)\end{aligned}$$

Solving for  $A$ :

$$\begin{aligned}\text{Let } x = 5 : &\implies 6(5)^2 + 7(5) - 3 = A(5^2 + 1) \\ &\implies 150 + 35 - 3 = 26A \\ &\implies 182 = 26A \\ &\implies A = 7\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}6x^2 + 7x - 3 &= 7(x^2 + 1) + Bx^2 - 5Bx + Cx - 5C \\ 6x^2 + 7x - 3 &= (7 + B)x^2 + (C - 5B)x + (7 - 5C)\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}7 + B &= 6 \\ \implies B &= -1\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}7 - 5C &= -3 \\ \implies -5C &= -10 \\ \implies C &= 2\end{aligned}$$

Thus,  $A = 7, B = -1, C = 2$ .

### Solution to Question 1ii:

Using the result from part (i):

$$\begin{aligned}\int \frac{6x^2 + 7x - 3}{(x - 5)(x^2 + 1)} dx &= \int \left( \frac{7}{x - 5} + \frac{-x + 2}{x^2 + 1} \right) dx \\ &= 7 \int \frac{1}{x - 5} dx - \int \frac{x}{x^2 + 1} dx + 2 \int \frac{1}{x^2 + 1} dx \\ &= 7 \ln |x - 5| - \frac{1}{2} \ln(x^2 + 1) + 2 \tan^{-1} x + C\end{aligned}$$

### Solution to Question 2i:

We set up the partial fraction decomposition:

$$\begin{aligned}\frac{15}{x^2 - 2x - 15} &= \frac{A}{x + 3} + \frac{B}{x - 5} \\ 15 &= A(x - 5) + B(x + 3)\end{aligned}$$

Solving for  $A$  and  $B$ :

$$\begin{aligned}\text{Let } x = 5 : & \implies 15 = B(5 + 3) \\ & \implies 15 = 8B \\ & \implies B = \frac{15}{8} \\ \text{Let } x = -3 : & \implies 15 = A(-3 - 5) \\ & \implies 15 = -8A \\ & \implies A = -\frac{15}{8}\end{aligned}$$

Thus,  $A = -\frac{15}{8}$  and  $B = \frac{15}{8}$ .

**Solution to Question 2ii:**

We need to find  $\int \frac{x^2-2x}{x^2-2x-15} dx$ . First, perform polynomial division since the degree of the numerator is equal to the degree of the denominator:

$$\begin{aligned}\frac{x^2 - 2x}{x^2 - 2x - 15} &= \frac{(x^2 - 2x - 15) + 15}{x^2 - 2x - 15} \\ &= 1 + \frac{15}{x^2 - 2x - 15}\end{aligned}$$

Now, integrate using the result from part (i):

$$\begin{aligned}\int \frac{x^2 - 2x}{x^2 - 2x - 15} dx &= \int \left( 1 + \frac{-15/8}{x+3} + \frac{15/8}{x-5} \right) dx \\ &= x - \frac{15}{8} \ln |x+3| + \frac{15}{8} \ln |x-5| + C \\ &= x + \frac{15}{8} \ln \left| \frac{x-5}{x+3} \right| + C\end{aligned}$$

**Solution to Question 3i:**

We set up the partial fraction decomposition:

$$\begin{aligned}\frac{5}{x^2(2-x)} &= \frac{ax+b}{x^2} + \frac{c}{2-x} \\ 5 &= (ax+b)(2-x) + cx^2\end{aligned}$$

Expanding the right side:

$$\begin{aligned}5 &= 2ax - ax^2 + 2b - bx + 2cx^2 - cx^2 \\ 5 &= (2c-a)x^2 + (2a-b)x + 2b\end{aligned}$$

Comparing coefficients:

Constant term:  $2b = 5$

$$\implies b = \frac{5}{2}$$

Coefficient of  $x$ :  $2a - b = 0$

$$\implies 2a = \frac{5}{2}$$

$$\implies a = \frac{5}{4}$$

Coefficient of  $x^2$ :  $2c - a = 0$

$$\implies 2c = \frac{5}{4}$$

$$\implies c = \frac{5}{8}$$

Thus,  $a = \frac{5}{4}, b = \frac{5}{2}, c = \frac{5}{8}$ .

**Solution to Question 3ii:**

We need to find  $\int \frac{20}{x^2(2-x)} dx$ . This is  $4 \times \int \frac{5}{x^2(2-x)} dx$ :

$$\begin{aligned} \int \frac{20}{x^2(2-x)} dx &= 4 \int \left( \frac{5/4x + 5/2}{x^2} + \frac{5/8}{2-x} \right) dx \\ &= 4 \int \left( \frac{5}{4x} + \frac{5}{2x^2} + \frac{5}{8(2-x)} \right) dx \\ &= 4 \left( \frac{5}{4} \ln|x| + \frac{5}{2} \cdot \frac{-1}{x} + \frac{5}{8} \cdot (-1) \ln|2-x| \right) + C \\ &= 5 \ln|x| - \frac{10}{x} - \frac{5}{2} \ln|2-x| + C \end{aligned}$$

**Solution to Question 4i:**

We set up the partial fraction decomposition:

$$\begin{aligned} \frac{3x^2 + 4x + 12}{x(x^2 + 4)} &= \frac{A}{x} + \frac{Bx + C}{x^2 + 4} \\ 3x^2 + 4x + 12 &= A(x^2 + 4) + (Bx + C)x \end{aligned}$$

Solving for  $A$ :

$$\begin{aligned} \text{Let } x = 0 : &\implies 12 = A(4) \\ &\implies A = 3 \end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned} 3x^2 + 4x + 12 &= 3x^2 + 12 + Bx^2 + Cx \\ 3x^2 + 4x + 12 &= (3 + B)x^2 + Cx + 12 \end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned} 3 + B &= 3 \\ \implies B &= 0 \end{aligned}$$

Comparing coefficients of  $x$ :

$$C = 4$$

Thus,  $A = 3, B = 0, C = 4$ .

**Solution to Question 4ii:**

Using the result from part (i):

$$\begin{aligned}\int \frac{3x^2 + 4x + 12}{x(x^2 + 4)} dx &= \int \left( \frac{3}{x} + \frac{4}{x^2 + 4} \right) dx \\ &= 3 \ln |x| + 4 \cdot \frac{1}{2} \tan^{-1} \left( \frac{x}{2} \right) + C \\ &= 3 \ln |x| + 2 \tan^{-1} \left( \frac{x}{2} \right) + C\end{aligned}$$

## Harder Exam Questions

**Solution to Question 1i:**

We start with the decomposition:

**Solution to Question 1ii:**

Using the result from part (i):

$$\begin{aligned}\int \frac{x^2}{4x^2 - 9} dx &= \int \left( \frac{1}{4} + \frac{3/8}{2x - 3} - \frac{3/8}{2x + 3} \right) dx \\ &= \frac{1}{4}x + \frac{3}{8} \cdot \frac{1}{2} \ln |2x - 3| - \frac{3}{8} \cdot \frac{1}{2} \ln |2x + 3| + C \\ &= \frac{1}{4}x + \frac{3}{16} \ln \left| \frac{2x - 3}{2x + 3} \right| + C\end{aligned}$$

**Solution to Question 2i:**

We set up the partial fraction decomposition:

$$\begin{aligned}\frac{-7x - 18}{(x - 1)(x^2 + 4)} &= \frac{A}{x - 1} + \frac{Bx + C}{x^2 + 4} \\ -7x - 18 &= A(x^2 + 4) + (Bx + C)(x - 1)\end{aligned}$$

Solving for  $A$ :

$$\begin{aligned}\text{Let } x = 1 : &\implies -7(1) - 18 = A(1^2 + 4) \\ &\implies -25 = 5A \\ &\implies A = -5\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}
-7x - 18 &= -5(x^2 + 4) + Bx^2 - Bx + Cx - C \\
-7x - 18 &= (-5 + B)x^2 + (C - B)x + (-20 - C)
\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}
-5 + B &= 0 \\
\implies B &= 5
\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}
-20 - C &= -18 \\
\implies -C &= 2 \\
\implies C &= -2
\end{aligned}$$

Thus,  $A = -5, B = 5, C = -2$ .

### Solution to Question 2ii:

Using the result from part (i):

$$\begin{aligned}
\int \frac{-7x - 18}{(x - 1)(x^2 + 4)} dx &= \int \left( \frac{-5}{x - 1} + \frac{5x - 2}{x^2 + 4} \right) dx \\
&= -5 \int \frac{1}{x - 1} dx + 5 \int \frac{x}{x^2 + 4} dx - 2 \int \frac{1}{x^2 + 4} dx \\
&= -5 \ln |x - 1| + \frac{5}{2} \ln(x^2 + 4) - 2 \cdot \frac{1}{2} \tan^{-1} \left( \frac{x}{2} \right) + C \\
&= -5 \ln |x - 1| + \frac{5}{2} \ln(x^2 + 4) - \tan^{-1} \left( \frac{x}{2} \right) + C
\end{aligned}$$

### Solution to Question 3i:

We set up the partial fraction decomposition:

$$\begin{aligned}
\frac{2x}{(x + 1)(x^2 + 1)} &= \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1} \\
2x &= A(x^2 + 1) + (Bx + C)(x + 1)
\end{aligned}$$

Solving for  $A$ :

$$\begin{aligned}
\text{Let } x = -1 : \implies 2(-1) &= A((-1)^2 + 1) \\
\implies -2 &= 2A \\
\implies A &= -1
\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}
2x &= -1(x^2 + 1) + Bx^2 + Bx + Cx + C \\
2x &= (-1 + B)x^2 + (B + C)x + (-1 + C)
\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}
-1 + B &= 0 \\
\implies B &= 1
\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}-1 + C &= 0 \\ \implies C &= 1\end{aligned}$$

Thus,  $A = -1, B = 1, C = 1$ .

**Solution to Question 3ii:**

Using the result from part (i):

$$\begin{aligned}\int_0^1 \frac{2x}{(x+1)(x^2+1)} dx &= \int_0^1 \left( \frac{-1}{x+1} + \frac{x+1}{x^2+1} \right) dx \\ &= \int_0^1 \left( \frac{-1}{x+1} + \frac{x}{x^2+1} + \frac{1}{x^2+1} \right) dx \\ &= \left[ -\ln|x+1| + \frac{1}{2} \ln(x^2+1) + \tan^{-1} x \right]_0^1 \\ &= \left( -\ln 2 + \frac{1}{2} \ln 2 + \tan^{-1} 1 \right) - \left( -\ln 1 + \frac{1}{2} \ln 1 + \tan^{-1} 0 \right) \\ &= -\frac{1}{2} \ln 2 + \frac{\pi}{4} + 0 \\ &= \frac{\pi}{4} - \frac{1}{2} \ln 2\end{aligned}$$

**Solution to Question 4:**

We decompose the integrand into partial fractions:

$$\begin{aligned}\frac{2x^2 + 5x + 9}{(x-1)(x^2 + 2x + 5)} &= \frac{A}{x-1} + \frac{Bx + C}{x^2 + 2x + 5} \\ 2x^2 + 5x + 9 &= A(x^2 + 2x + 5) + (Bx + C)(x-1)\end{aligned}$$

Solving for  $A$ :

$$\begin{aligned}\text{Let } x = 1 : \implies 2(1)^2 + 5(1) + 9 &= A(1^2 + 2(1) + 5) \\ \implies 16 &= 8A \\ \implies A &= 2\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}2x^2 + 5x + 9 &= 2(x^2 + 2x + 5) + Bx^2 - Bx + Cx - C \\ 2x^2 + 5x + 9 &= (2+B)x^2 + (4-B+C)x + (10-C)\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}2 + B &= 2 \\ \implies B &= 0\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}10 - C &= 9 \\ \implies C &= 1\end{aligned}$$

Now, integrate:

$$\begin{aligned}\int \frac{2x^2 + 5x + 9}{(x-1)(x^2 + 2x + 5)} dx &= \int \left( \frac{2}{x-1} + \frac{1}{x^2 + 2x + 5} \right) dx \\ &= 2 \ln |x-1| + \int \frac{1}{(x+1)^2 + 4} dx \\ &= 2 \ln |x-1| + \frac{1}{2} \tan^{-1} \left( \frac{x+1}{2} \right) + C\end{aligned}$$

**Solution to Question 5i:**

We are asked to show that  $\frac{1}{x(1+x^2)} = \frac{1}{x} - \frac{x}{1+x^2}$ . We can do this by combining the fractions on the right side:

$$\begin{aligned}\frac{1}{x} - \frac{x}{1+x^2} &= \frac{(1+x^2) - x(x)}{x(1+x^2)} \\ &= \frac{1+x^2-x^2}{x(1+x^2)} \\ &= \frac{1}{x(1+x^2)}\end{aligned}$$

Thus, the identity is shown.

**Solution to Question 5ii:**

We need to find  $\int \frac{1}{x^2} \tan^{-1} x dx$ . We use integration by parts:  $\int u dv = uv - \int v du$ . Let  $u = \tan^{-1} x$  and  $dv = \frac{1}{x^2} dx$ . Then  $du = \frac{1}{1+x^2} dx$  and  $v = -\frac{1}{x}$ .

$$\begin{aligned}\int \frac{1}{x^2} \tan^{-1} x dx &= (\tan^{-1} x) \left( -\frac{1}{x} \right) - \int \left( -\frac{1}{x} \right) \frac{1}{1+x^2} dx \\ &= -\frac{\tan^{-1} x}{x} + \int \frac{1}{x(1+x^2)} dx\end{aligned}$$

Using the result from part (i):

$$\begin{aligned}\int \frac{1}{x(1+x^2)} dx &= \int \left( \frac{1}{x} - \frac{x}{1+x^2} \right) dx \\ &= \ln |x| - \frac{1}{2} \ln(1+x^2) + C\end{aligned}$$

Substituting this back into the integration by parts result:

$$\int \frac{1}{x^2} \tan^{-1} x dx = -\frac{\tan^{-1} x}{x} + \ln |x| - \frac{1}{2} \ln(1+x^2) + C$$

**Solution to Question 6i:**

We set up the partial fraction decomposition:

$$\begin{aligned}\frac{5-5x^2}{(1+2x)(1+x^2)} &= \frac{a}{1+2x} + \frac{bx+c}{1+x^2} \\ 5-5x^2 &= a(1+x^2) + (bx+c)(1+2x)\end{aligned}$$

Solving for  $a$ :

$$\begin{aligned}\text{Let } x = -1/2 : &\implies 5 - 5(-1/2)^2 = a(1 + (-1/2)^2) \\ &\implies 5 - 5/4 = a(5/4) \\ &\implies 15/4 = 5/4a \\ &\implies a = 3\end{aligned}$$

Expanding and comparing coefficients:

$$\begin{aligned}5 - 5x^2 &= 3(1 + x^2) + bx + 2bx^2 + c + 2cx \\ 5 - 5x^2 &= (3 + 2b)x^2 + (b + 2c)x + (3 + c)\end{aligned}$$

Comparing coefficients of  $x^2$ :

$$\begin{aligned}3 + 2b &= -5 \\ \implies 2b &= -8 \\ \implies b &= -4\end{aligned}$$

Comparing coefficients of  $x^0$ :

$$\begin{aligned}3 + c &= 5 \\ \implies c &= 2\end{aligned}$$

Thus,  $a = 3, b = -4, c = 2$ .

### **Solution to Question 6ii:**

Using the result from part (i):

$$\begin{aligned}\int_0^1 \frac{5 - 5x^2}{(1 + 2x)(1 + x^2)} dx &= \int_0^1 \left( \frac{3}{1 + 2x} + \frac{-4x + 2}{1 + x^2} \right) dx \\ &= \int_0^1 \frac{3}{1 + 2x} dx - 4 \int_0^1 \frac{x}{1 + x^2} dx + 2 \int_0^1 \frac{1}{1 + x^2} dx \\ &= \left[ \frac{3}{2} \ln |1 + 2x| - 2 \ln(1 + x^2) + 2 \tan^{-1} x \right]_0^1 \\ &= \left( \frac{3}{2} \ln 3 - 2 \ln 2 + 2 \tan^{-1} 1 \right) - \left( \frac{3}{2} \ln 1 - 2 \ln 1 + 2 \tan^{-1} 0 \right) \\ &= \frac{3}{2} \ln 3 - 2 \ln 2 + 2 \frac{\pi}{4} \\ &= \frac{3}{2} \ln 3 - 2 \ln 2 + \frac{\pi}{2} \\ &= \frac{1}{2} (3 \ln 3 - 4 \ln 2 + \pi) \\ &= \frac{1}{2} (\ln 27 - \ln 16 + \pi) \\ &= \frac{1}{2} \left( \pi + \ln \frac{27}{16} \right)\end{aligned}$$

### **Solution to Question 6iii:**

We use the substitution  $t = \tan x$ . Then  $dt = \sec^2 x dx = (1 + t^2) dx$ , so  $dx = \frac{dt}{1+t^2}$ . Also,  $\cos 2x = \frac{1-t^2}{1+t^2}$  and  $\sin 2x = \frac{2t}{1+t^2}$ .

$$\begin{aligned}
\int_0^{\pi/4} \frac{\cos 2x}{1 + 2 \sin 2x + \cos 2x} dx &= \int_0^1 \frac{\frac{1-t^2}{1+t^2}}{1 + 2 \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{dt}{1+t^2} \\
&= \int_0^1 \frac{1-t^2}{(1+t^2) + 4t + (1-t^2)} \cdot \frac{1}{1+t^2} dt \\
&= \int_0^1 \frac{1-t^2}{2+4t} \cdot \frac{1}{1+t^2} dt \\
&= \int_0^1 \frac{1-t^2}{2(1+2t)(1+t^2)} dt \\
&= \frac{1}{2} \int_0^1 \frac{1-t^2}{(1+2t)(1+t^2)} dt
\end{aligned}$$

Now, we notice that the integrand is a multiple of the one in part (ii). Specifically,  $\frac{1-t^2}{(1+2t)(1+t^2)} = \frac{1}{5} \frac{5-5t^2}{(1+2t)(1+t^2)}$ .

$$\begin{aligned}
\int_0^{\pi/4} \frac{\cos 2x}{1 + 2 \sin 2x + \cos 2x} dx &= \frac{1}{2} \cdot \frac{1}{5} \int_0^1 \frac{5-5t^2}{(1+2t)(1+t^2)} dt \\
&= \frac{1}{10} \cdot \frac{1}{2} \left( \pi + \ln \frac{27}{16} \right) \\
&= \frac{1}{20} \left( \pi + \ln \frac{27}{16} \right)
\end{aligned}$$